

IDEATION STAGE FOR ROBOT MAKING COMPETITION
ROBOFEST GUJARAT-5.0

Application No.:

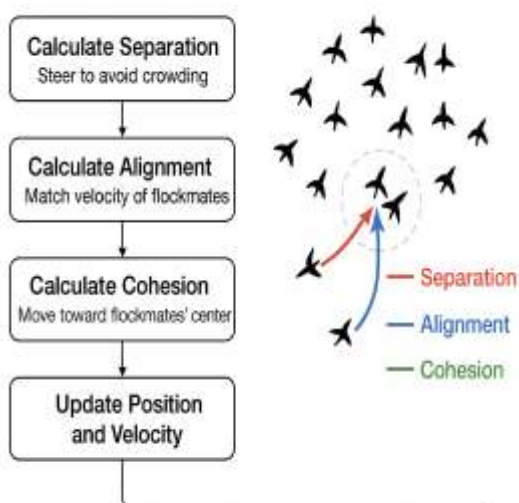
TECHNICAL DETAILS FOR PROPOSED ROBOT

(Wherever necessary separate sheet/page is allowed to attach; Institute may submit extra details if find necessary)

1. Type of Robot:

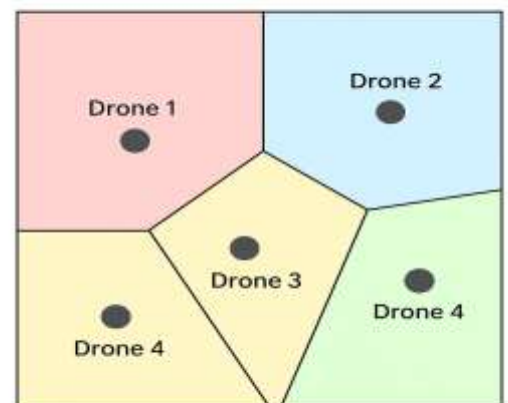
The proposed robot is a multi-agent system known as "Project S.C.A.N. (Swarm Coordinated Autonomous Navigation)," a novel and cost-effective framework designed for human-directed swarm navigation. The system consists of a swarm of lightweight Unmanned Aerial Vehicles (UAVs), each weighing less than 454 grams, tasked with autonomously surveying a 100-meter minefield. The primary objective is to identify and map the location of both surface and sub-surface mines and, based on this information, guide a human operator along a safe path within a strict 10-minute timeframe.

The core of the system is its fully decentralized architecture. This design philosophy is a direct response to the project's most significant constraint: the mandate for "ONLY onboard computing," which prohibits any reliance on a powerful ground station or cloud server for computationally intensive tasks. As such, each drone in the swarm is a fully autonomous agent, capable of individual sensing, peer-to-peer communication, collective planning, and independent action. This decentralized paradigm is a deliberate choice that not only adheres to the competition's rules but also creates a system that is inherently more robust and scalable, as the failure of a single drone does not compromise the entire mission. This design aligns with cutting-edge research in robotics and demonstrates a bio-inspired approach to problem-solving, where complex, intelligent behaviour emerges from a collection of simple, locally governed interactions.



Craig Reynolds Classic Boids

**Voronoi Tessellation
for Drone Task Allocation**



Minefield

2. Robot Assembly Design (Proposed Diagram): Drawings each part of the robot is preferred as an attachment. (CAD drawings are preferred) (400 words)

The physical architecture of each S.C.A.N. drone is meticulously designed to create a lightweight, robust, and aerodynamically stable platform. The assembly, visualized in the provided CAD drawings, prioritizes a low centre of gravity, strategic component placement to minimize electronic interference, and a modular design for ease of maintenance and rapid prototyping. Every component is positioned to maximize performance while adhering strictly to the sub-450-gram weight constraint.

a) Frame and Core Structure:

The foundation of the drone is a custom-designed **X-frame**, a configuration renowned for its exceptional manoeuvrability and stability. The frame will be constructed from either 2mm **carbon fibre** plates for maximum rigidity and minimal weight or a durable, 3D-printed polymer like **PETG** for rapid design iteration. The key dimensions are:

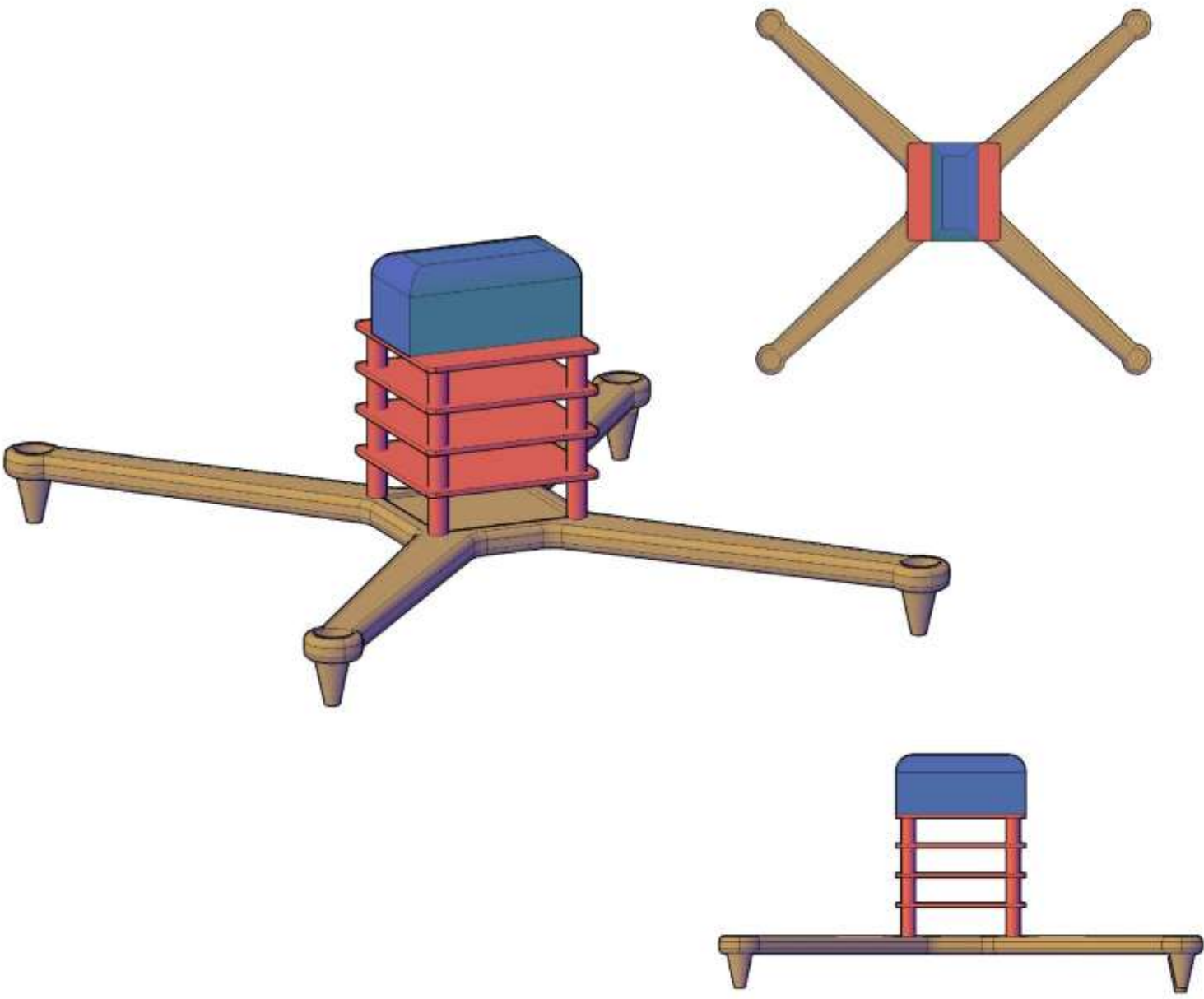
- **Motor-to-Motor length:** 31.6 cm
- **Motor-to-Motor breadth:** 27.5 cm
- **Frame Angle (Short Axis):** 82 °
- **Vertical Height:** 11 cm

This stretched-X geometry provides more stable pitch control and isolates the forward-facing camera and sensors from being obscured by the propellers.

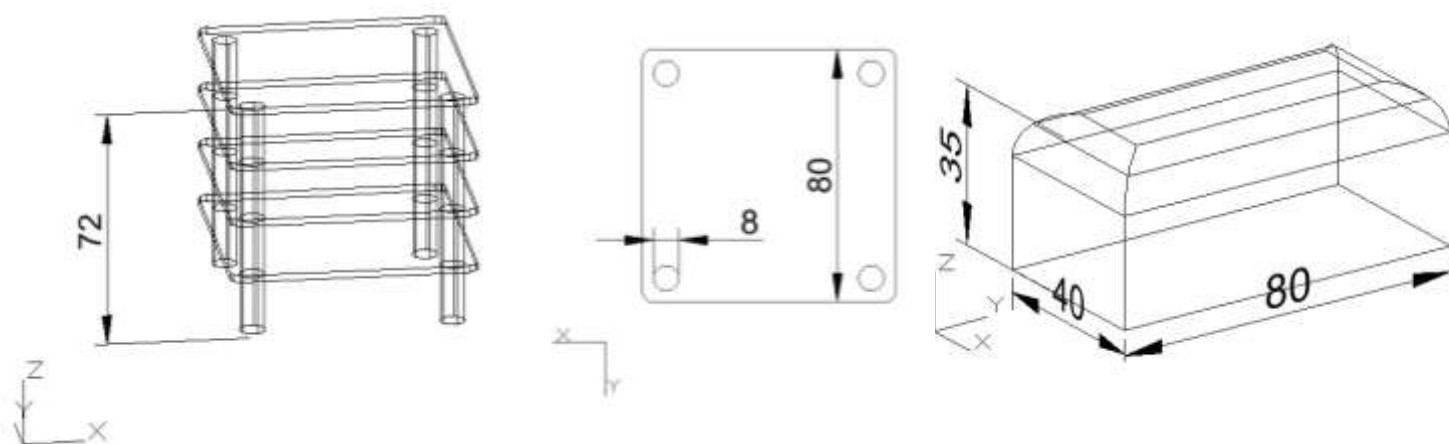
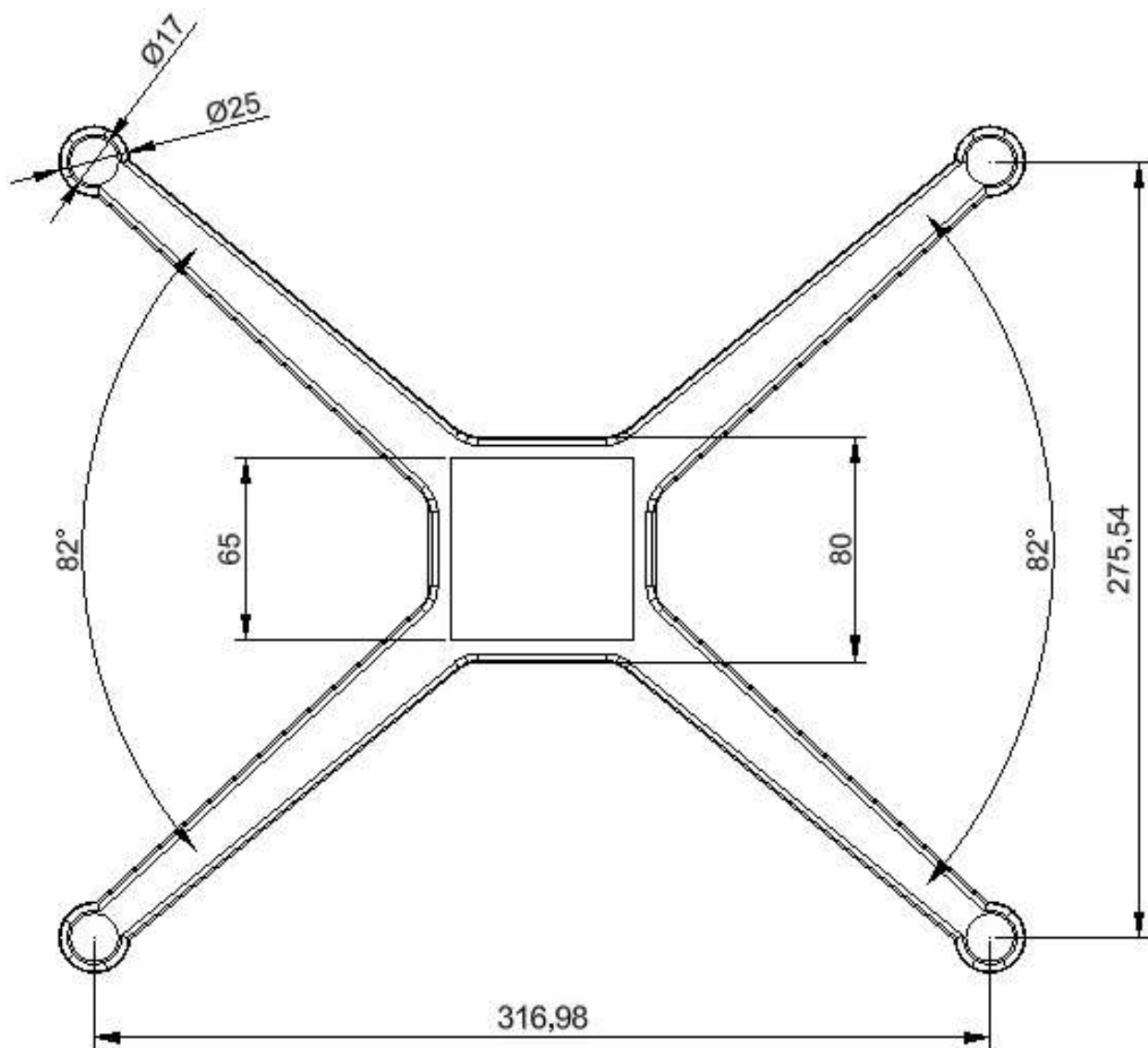
b) Component Layout and Integration:

The components are arranged in a dense, centrally located stack to concentrate the mass and ensure a balanced centre of gravity, which is critical for precise flight control.

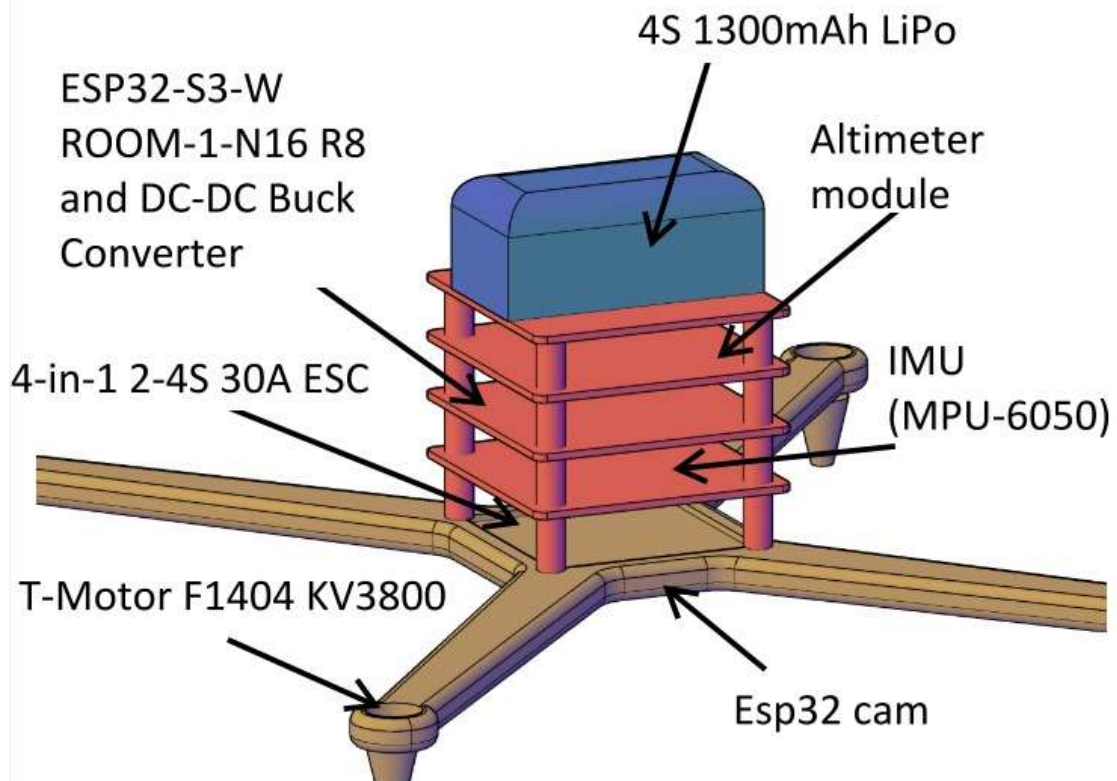
- **Bottom Plate:** This plate serves as the mounting point for the **4-in-1 ESC**, which is the first component in the electronics stack. Directly beneath this plate, a custom 3D-printed housing holds the downward-facing **ToF sensor** for mine detection, ensuring an unobstructed view of the ground.
- **Main Plate (Central Hub):** Positioned directly above the ESC, the **ESP32-S3 microcontroller** sits at the heart of the drone. The **IMU** is mounted as close to the centre of this board as possible to provide the most accurate rotational data. The **Buck Converter** is also placed on this level, situated away from sensitive sensors to prevent electromagnetic interference.
- **Top Plate and Sensor Suite:** The top plate provides a mounting platform for the **4S LiPo Battery**, secured with a Velcro strap for easy removal. The forward-facing **ToF sensors** for obstacle avoidance are mounted on the front of this plate, while the **INMP441 Microphone** is placed on a small, vibration-dampened stalk to isolate it from motor noise and ensure clear audio capture. The **Altimeter** is also housed here, away from the propeller wash to get accurate barometric pressure readings. The entire electronics stack is protected by a lightweight 3D-printed "cage" that provides impact resistance without significantly adding to the overall weight. This thoughtful, layered assembly ensures that each component functions optimally within a compact and resilient structure.



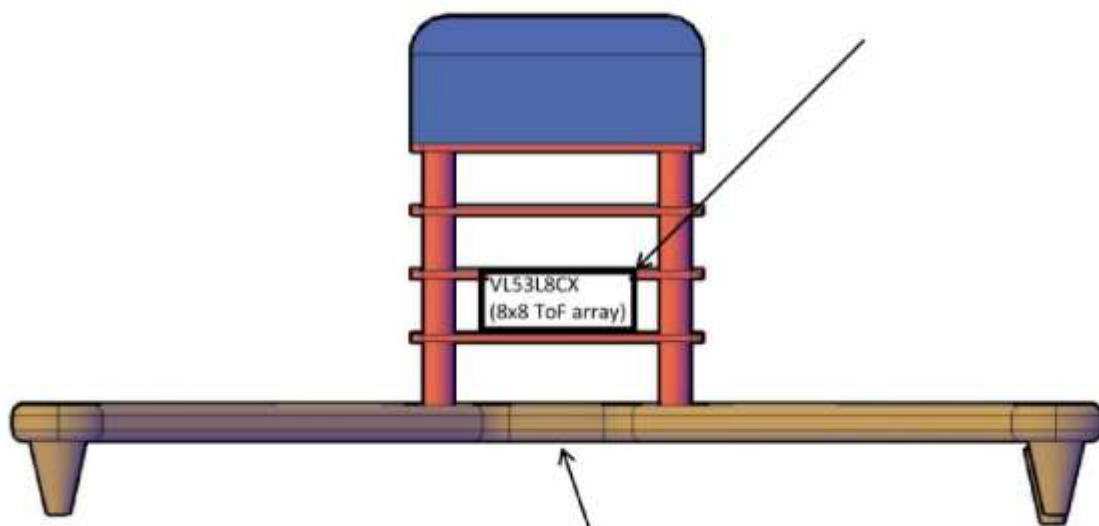
(The above model is made using AutoCAD)



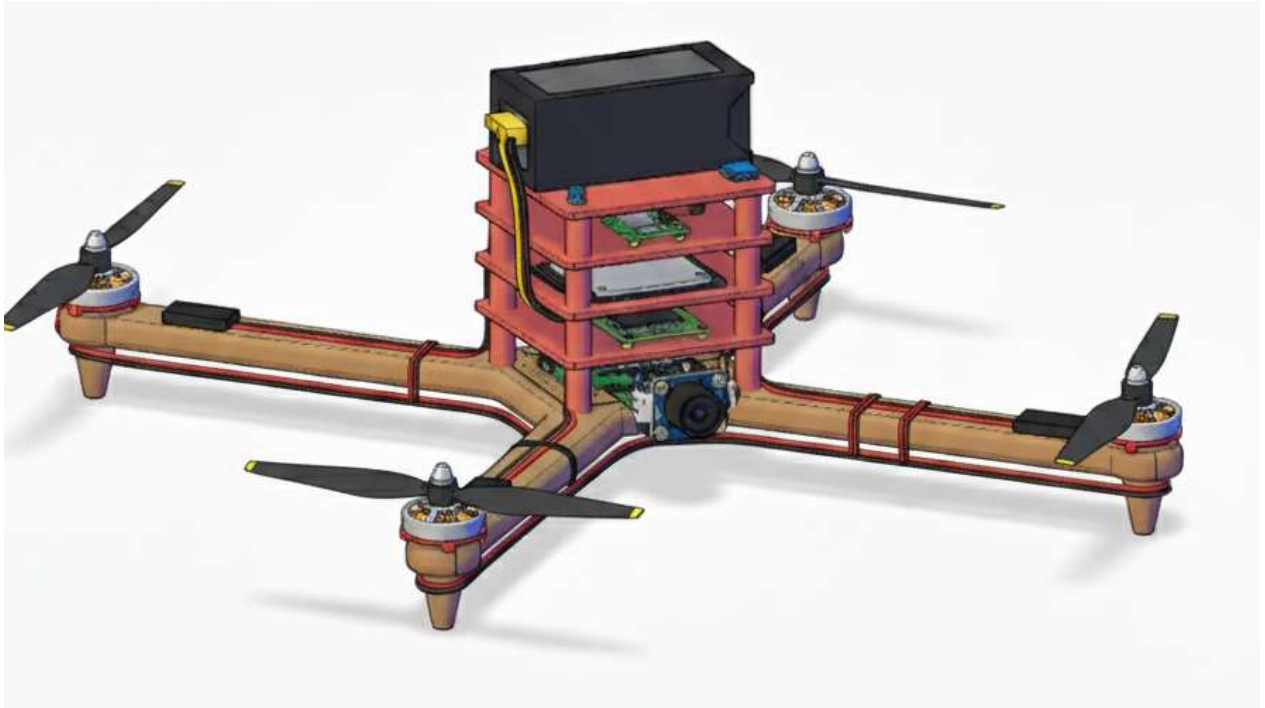
(All the measurements are in mm)



Four VL53L8CX attached on four lateral sides to provide obstacle avoidance and safe distance maintenance .



One VL53L8CX for mine field detection



(The above illustration is regenerated using AI(Gemini))

[Link to our CAD Designs](#)

3. Components to be used: (Enlist all the components with their make/company in four groups as enlisted in the following table:

- a. List of Structure components: like beams, bushes, shafts, belts, plates, pins, pulleys, wheel, connectors, batteries, motors, etc.

Photo	Name	Description	Quantity	Total Price /Buying Link
	4S 1300mAh LiPo	High energy density and discharge rate for performance and >10 min endurance.	1	₹3000
	T-Motor F1404 KV3800	The 1404 motor required flight time for lightweight drones, avoiding strain and ensuring reliable coordination.	4	₹6400
	GEMFAN 3016 HURRICANE	Offers efficient, low-noise flight with minimal vibration, precise control, and proven compatibility for 1404 motors on 4S	4	₹600

- b. List of Electronics Components: like Smart ports, switches, joysticks, controllers, LED/LCD screen, power supply, programming components, etc.

Photo	Name	Description	Quantity	Price/Buying link
	ESP32-S3-WROOM-1-N16R8	Dual-core, vector instructions for AI, integrated Wi-Fi/BT, ESP-NOW support.	1	₹700
	DC-DC Buck Converter	Steps down 4S LiPo voltage to a stable 5V for the MCU and sensors.	1	₹120
	4-in-1 2-4S 30A ESC	Controls brushless motors, compatible with 4S LiPo voltage.	1	₹4000
	BME280	Provides precise barometric pressure data, perfect for accurate altitude hold and environmental sensing	1	₹290
	MPU-6050	6-DoF Accelerometer + Gyroscope. Standard for DIY drones, sufficient for stabilization.	1	₹170
	VL53L8CX	8x8 ToF array. Enables high-quality, low-power C-SLAM and mine detection.	5	₹9000
	INMP441	A high-quality, omnidirectional MEMS microphone with a direct I2S digital interface for clear audio.	1	₹200

Component Weight Table

<i>Component</i>	<i>Approximate Weight (grams)</i>
Drone Frame (Custom made)	200 g
Microcontroller (ESP32-WROOM-32)	5 g
IMU (MPU-6050)	1 g
Barometer (BME280)	1.2 g
Optical/IR Sensor (VL53L8CX)	2.5 g
Camera (ESP32-CAM)	8.8 g
Microphone (INMP441)	1.6 g
Brushless Motors (4x)	40 g
Battery & ESCs	100 g
Total Weight per Drone	~ 360.1 g

4. The methodology of Making Robot: (Please write Entire Technical Specifications of Proposed Robot with brief notes and diagrams) (400 words)

Project S.C.A.N. is built on a decentralized architecture that treats onboard-only computing not as a limitation, but as a design advantage. This approach eliminates single points of failure and enables scalable, resilient swarm behaviour inspired by natural systems. The methodology is structured around a Tri-Pillar framework to address the real-time demands of autonomous minefield mapping.

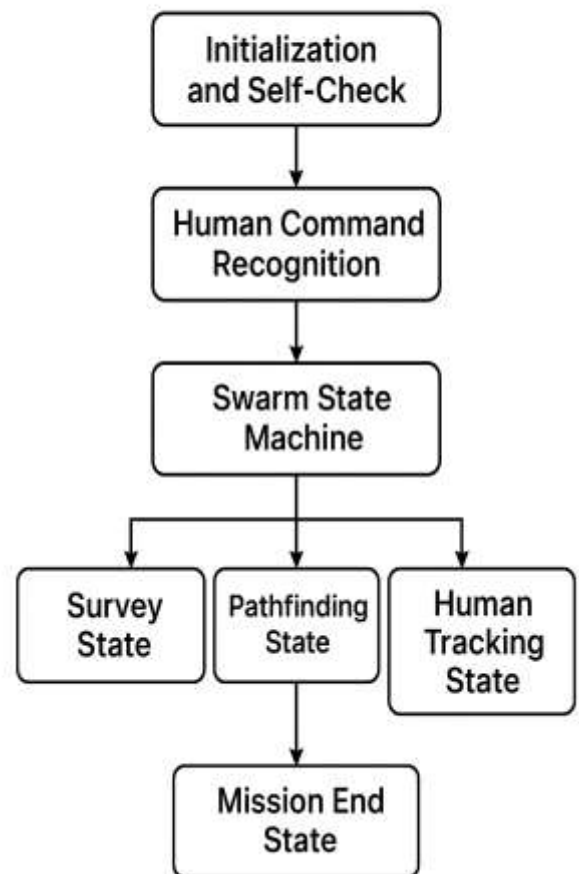
The first pillar, **Human-Swarm Interface (HSI)**, bridges the human operator and the robotic swarm. Using the ESP32-S3 microcontroller's vector instructions, the system runs lightweight neural networks for gesture and voice recognition. A key innovation is the development of a *Swarm Interaction Grammar*—a formal language that translates simple commands (e.g., "SURVEY AHEAD") into coordinated swarm actions. This reduces operator cognitive load from $O(n)$ to $O(1)$, enabling intuitive control of multiple agents.

The second pillar, **Decentralized Swarm Intelligence (DSI)**, governs inter-drone communication via **ESP-NOW**, a low-latency, peer-to-peer protocol requiring no external hardware. Movement coordination is based on bio-inspired algorithms like Boids, using rules of separation, alignment, and cohesion. For efficient minefield coverage, a decentralized task allocation strategy based on **Voronoi Tessellation** dynamically assigns regions to each drone, ensuring minimal overlap and full area coverage without central control.

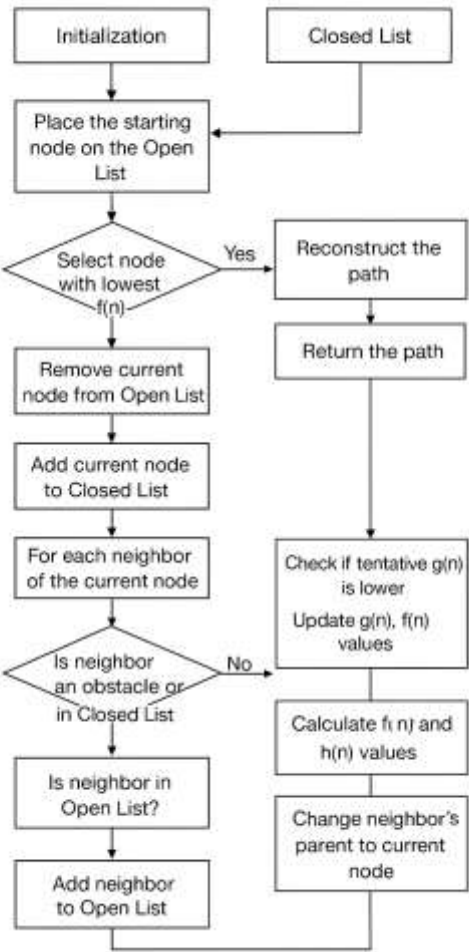
The third pillar, **Collaborative Perception & Planning (CPP)**, enables the swarm to build a shared understanding of the environment. Each drone uses an ultra-lightweight Collaborative SLAM (C-SLAM) algorithm optimized for low-resolution depth data from VL53L8CX Time-of-Flight sensors. Local maps are fused through inter-drone loop closures to create a unified probabilistic occupancy grid. This map supports global path planning for the human operator using A^* , while drones perform local navigation with RRT* to avoid dynamic obstacles and other agents.

Integrating these modules on a single ESP32-S3 presents a significant engineering challenge. A Real-Time Operating System (RTOS) is used to manage concurrent tasks with priority scheduling across dual CPU cores. The project follows a phased implementation—starting with single-drone validation, then two-drone integration, and finally full swarm demonstration. This staged approach ensures subsystem reliability and mitigates integration risks.

Proposed Logic for a Symmetrical Swarm



Overall, Project S.C.A.N. demonstrates a practical and scalable methodology for autonomous swarm robotics, emphasizing robust coordination, efficient sensing, and real-time decision-making under resource constraints.

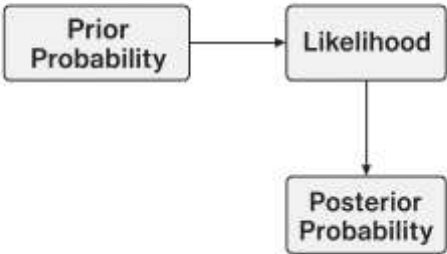


A* Algorithm



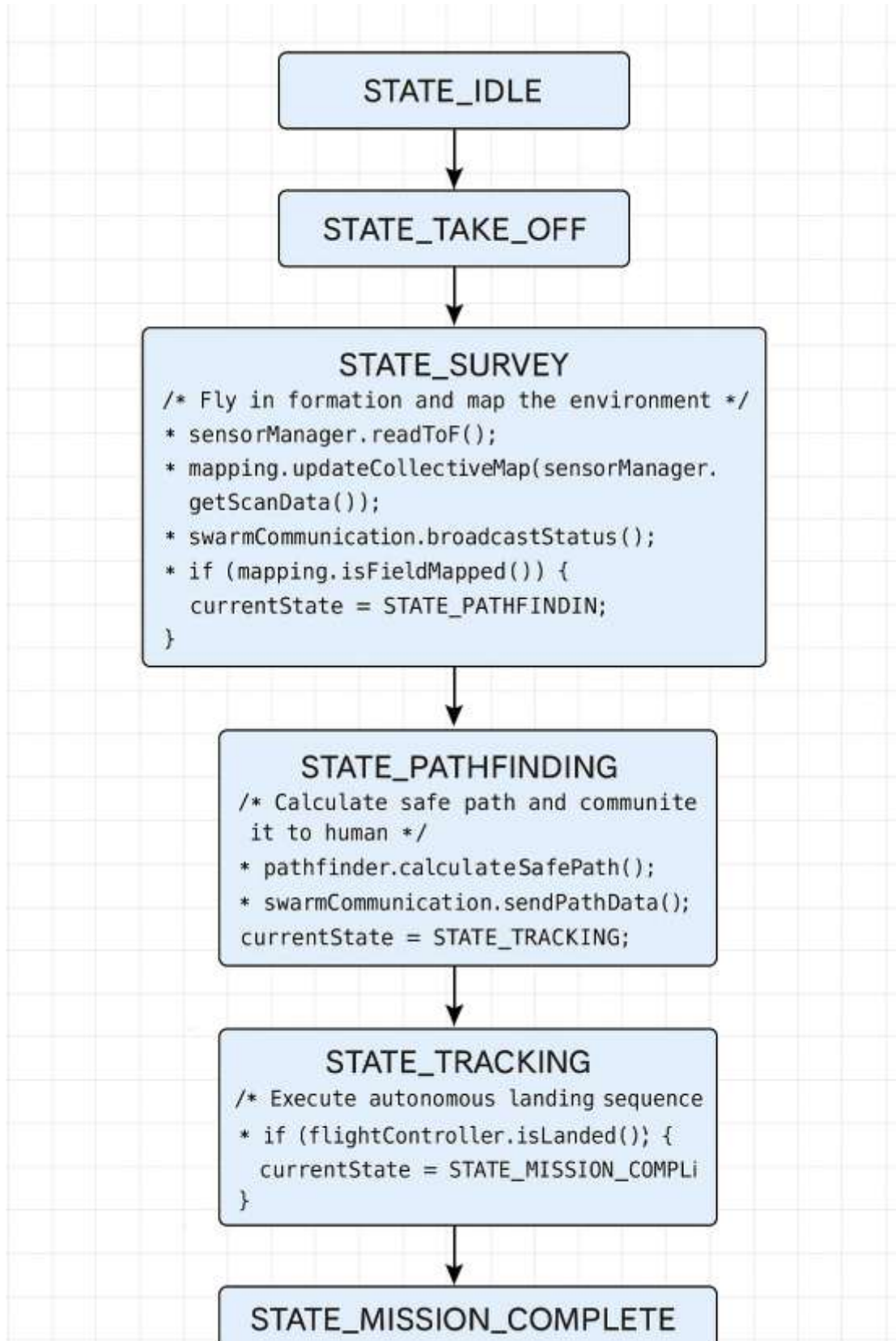
C-Slam Representation

$$P(H | E) = \frac{P(E | H) \cdot P(H)}{P(E)}$$



Bayesian Inference (For Predicting Mines)

CODE ALGORITHM



Swarm Interaction Grammar: Command Primitives

Command Type	User Action / Spoken Command	Interpreted Command	Resulting Swarm Action
Gesture	Open Palm, Push Forward	SURVEY, FORWARD	The swarm initiates a coordinated search pattern (e.g., line abreast) in the indicated direction.
Gesture	Fist, Point at Ground	FOCUS, POINTED_AREA	The swarm converges on the designated area, tightening its search pattern for higher-resolution mapping.
Gesture	"Come Hither" Motion	REPORT	The swarm returns to a holding pattern near the operator to await further commands.
Gesture	Two Fingers, Swipe on Ground	SHOW_PATH	The designated "pathfinder" drone begins visually indicating the determined safe route.
Voice	"SURVEY AHEAD"	SURVEY, FORWARD	Equivalent to the "Open Palm, Push Forward" gesture; initiates a forward survey.
Voice	"MARK SAFE PATH"	SHOW_PATH	Equivalent to the "Two Fingers, Swipe" gesture; commands the drone to mark the path.
Voice	"HOLD POSITION"	HOLD	The swarm immediately halts its current task and hovers in its present location.
Voice	"NEXT"	NEXT	Swarm re-arranges and aligns to show the next straightest portion of the safe path

A robust system should support both modalities. A vision-based gesture system is intuitive and silent, while a voice-based system is effective even when the operator's hands are occupied.

5. Feedback Mechanisms: Communicating the Safe Path

The final and most critical phase of the Human-Swarm Interaction loop is communicating the computed safe path to the operator. This is achieved through a dual-guidance system: physical drone alignment for immediate direction and a real-time map on a screen for overall situational awareness.

Initially, the drones will align to form the longest, safest, straight-line path from the operator's starting point. The operator's screen will display the complete minefield map, their current location, and this highlighted path.

The operator follows the physical drones. Once they reach the end of the marked segment, they give the voice command, **"NEXT."**

Instantly, two things happen:

1. The drone swarm re-calculates and physically aligns to show the next straightest portion of the safe path from the operator's new position.
2. The map on the operator's screen updates, showing their new current location and the next highlighted path segment ahead.

This simple "walk, command, and update" cycle provides clear, interactive, and reliable guidance through the entire minefield.

6. Application of proposed Robot in a societal context: (Not more than 100 words)

Beyond the immediate goal of the robotics competition, the technology developed for Project S.C.A.N. possesses significant potential for real-world applications with substantial societal benefits. The core capabilities of autonomous navigation in unknown environments, collaborative mapping, and intuitive human-swarm interaction are directly transferable to scenarios where human personnel face danger or inefficiency.

- **Search and Rescue (SAR):**
Swarm drones can safely survey disaster zones, detecting heat signatures or signs of life without risking human personnel.
- **Infrastructure Inspection:**
Enables multi-angle assessment of bridges, pipelines, and power lines, reducing the need for hazardous manual inspections.
- **Environmental Monitoring:**
Lightweight drones can track wildlife, assess vegetation health, and detect wildfires non-invasively across large areas.
- **Scalable & Resilient Design:**
Low-cost, decentralized architecture ensures adaptability in dynamic, high-risk environments.
- **Humanitarian & Industrial Impact:**
Core technologies support safer, more efficient operations in critical real-world applications.

7. Size of robot proposed for Proof of concept (Small version):

- a. Length in cm: ~30cm b. Width in cm: ~30cm c. Height in cm: ~ 10cm

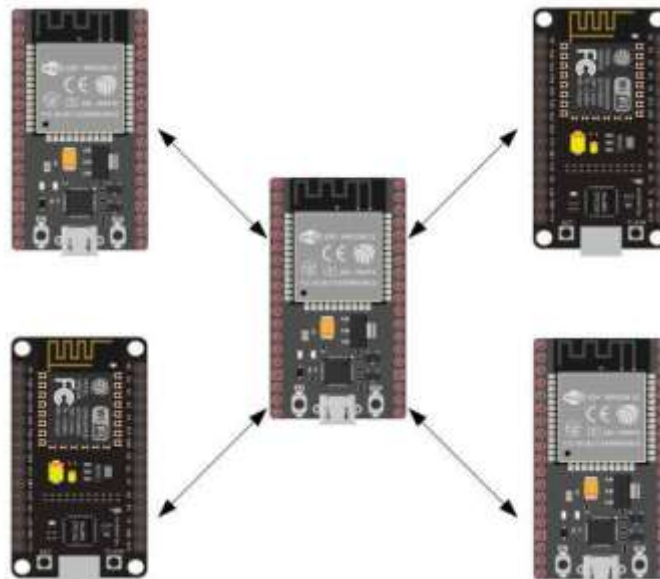
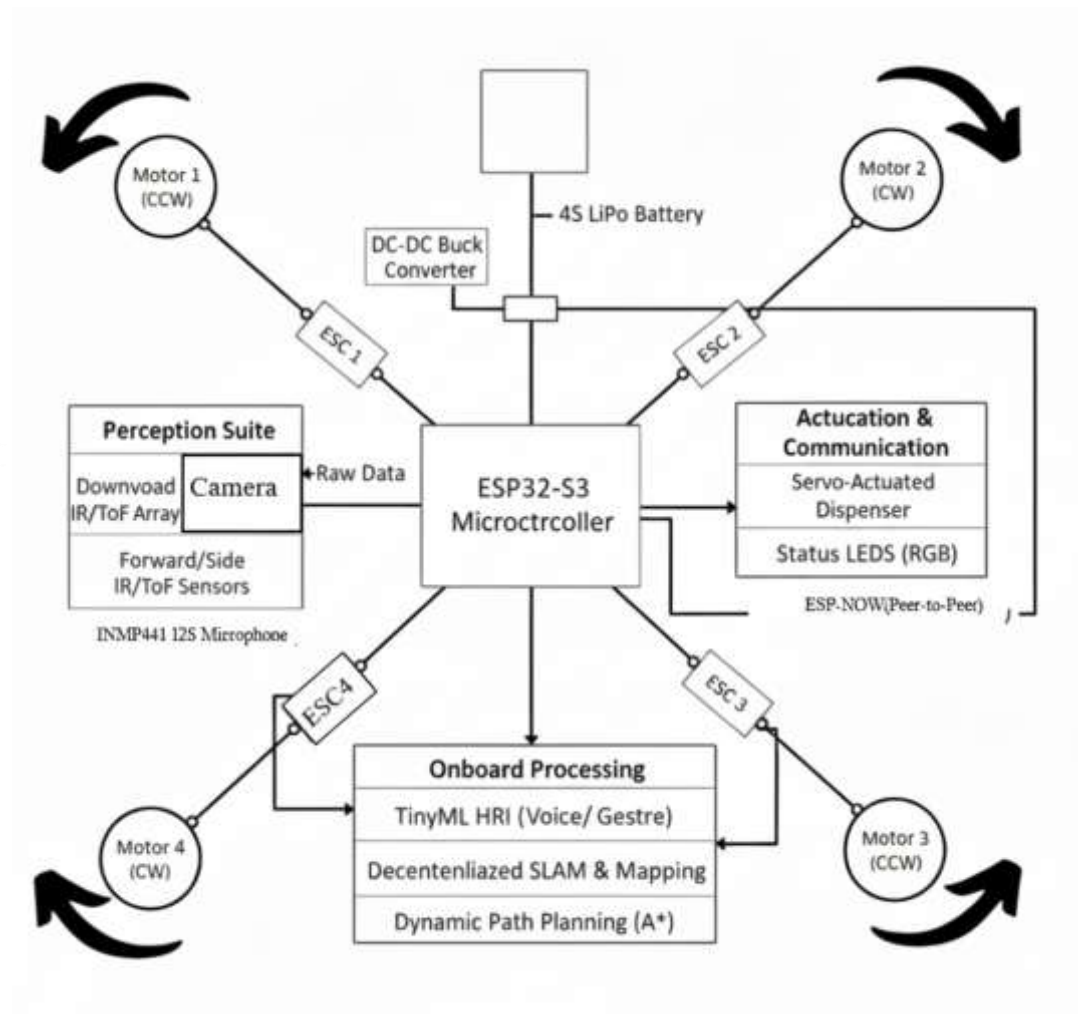
8. Size of robot proposed as Proto type (Actual Version)

- a. Length in cm: ~30cm b. Width in cm: ~ 27cm c. Height in cm: ~10cm

9. Timeline for Robot making with milestones. (Divided in activities vs. no. of days)

[illegible][illegible]

10. Please attach the proposed outline (photograph) for understanding of the evaluation committee.



ESP-NOW Communication Protocol

11. The Path to Innovation: Identifying Opportunities for Novelty.

To create a patentable and impactful technology, this project must innovate beyond merely integrating existing open-source components. Our strategy focuses on three core areas of innovation:

- **Novel, Ultra-Lightweight Collaborative SLAM:** We will develop a C-SLAM algorithm specifically optimized for the severe computational and memory limitations of nano-drones. This involves creating a system that works with low-resolution depth sensors and a bandwidth-efficient communication protocol that intelligently exchanges compressed map data.
- **Formalized "Swarm Interaction Grammar":** The key innovation is not just gesture or voice recognition, but the creation of a formal grammar that allows simple, intuitive commands to be combined into complex, high-level swarm tactics. This elevates the human-robot interaction from direct control to supervisory oversight.
- **Holistic System Integration on a Resource-Constrained Platform:** The ultimate novelty is the successful, simultaneous execution of real-time Human-Swarm Interaction (HSI), decentralized coordination, and collaborative SLAM on a platform built entirely from low-cost, off-the-shelf microcontrollers and sensors. This vertical integration will set a new benchmark for what is possible in resource-constrained autonomous systems.

The most defensible intellectual property will not come from a single algorithm, but from the synergistic co-design of the entire system. While patenting a specific SLAM algorithm is difficult, the true innovation of Project S.C.A.N. is the unique architectural solution that enables HSI, swarm intelligence, and collaborative mapping to function together in real-time on an ESP32. A patent claim for this integrated system architecture—the "how" rather than the "what"—is far more powerful and protects the core of our innovation.

12. References

- 1) Deployable Swarms: Decentralized Multi-Agent Path Planning and Control | NIMBUS Lab, accessed August 18, 2025, <https://nimbus.unl.edu/deployable-swarms-decentralized-multi-agent-path-planning-and-control/>
- 2) UAVs Meet LLMs: Overviews and Perspectives Toward Agentic Low-Altitude Mobility - arXiv, accessed August 18, 2025, <https://arxiv.org/html/2501.02341v1>
- 3) Human Swarm Interaction - Technische Hochschule Ingolstadt, accessed August 18, 2025, <https://www.thi.de/forschung/technologietransferzentrum-unbemannte-flugsysteme/forschungsbereiche/projekte-im-ttz/human-swarm-interaction/>
- 4) Human-Swarm Interaction with a Gesture-Controlled Aerial Robot Formation for Safety Monitoring Applications, accessed August 18, 2025, <https://aiia.csd.auth.gr/wp-content/uploads/2023/11/UAVFormationControlWithGesturess.pdf>
- 5) How Drones Are Revolutionizing Search and Rescue - IEEE Public Safety Technology, accessed August 18, 2025, <https://publicsafety.ieee.org/topics/how-drones-are-revolutionizing-search-and-rescue/>
- 6) Mission Chain Driven Unmanned Aerial Vehicle Swarms Cooperation for the Search and Rescue of Outdoor Injured Human Targets - MDPI, accessed August 18, 2025, <https://www.mdpi.com/2504-446X/6/6/138>
- 7) DVM-SLAM: Decentralized Visual Monocular Simultaneous Localization and Mapping for Multi-Agent Systems - arXiv, accessed August 18, 2025, <https://arxiv.org/html/2503.04126v1>
- 8) Multi-UAV Collaborative Monocular SLAM - Research Collection, accessed August 18, 2025, <https://www.research-collection.ethz.ch/bitstream/handle/20.500.11850/272499/eth-50606-01.pdf>
- 9) Speech Recognition - RobotShop, accessed August 18, 2025, <https://www.robotshop.com/collections/speech-recognition>
- 10) ETH-PBL/Nano_Swarm_Mapping: Fully Onboard SLAM for Distributed Mapping with a Swarm of Nano-Drones - GitHub, accessed August 18, 2025, https://github.com/ETH-PBL/Nano_Swarm_Mapping

- 11) A Path-Planning Method for UAV Swarm under Multiple Environmental Threats - MDPI, accessed August 18, 2025, <https://www.mdpi.com/2504-446X/8/5/171>
- 12) Multi-Agent Reinforcement Learning for Coordinated Drone Swarms - ijarie, accessed August 18, 2025, https://ijariie.com/AdminUploadPdf/Multi_Agent_Reinforcement_Learning_for_Coordinated_Drone_Swarms_ijariie26396.pdf
- 13) Solving Multi-Agent Safe Optimal Control with Distributed Epigraph Form MARL - Robotics, accessed August 18, 2025, <https://www.roboticsproceedings.org/rss21/p027.pdf>
- 14) A Multi-Label A* Algorithm for Multi-Agent Pathfinding - andrew.cmu.edu, accessed August 24, 2025, https://www.andrew.cmu.edu/user/vanhoeve/papers/MAPD_ICAPS_2019.pdf
- 15) Fully Onboard SLAM for Distributed Mapping with a Swarm of Nano-Drones - ResearchGate, accessed August 18, 2025, https://www.researchgate.net/publication/373754415_Fully_Onboard_SLAM_for_Distributed_Mapping_with_a_Swarm_of_Nano-Drones/download
- 16) TensorFlow Lite for microcontrollers, accessed August 18, 2025, <https://cms.tinyml.org/wp-content/uploads/2020/09/bay-area-situnayake.pdf>
- 17) DRDO Develops Air-Launched Anti-Radiation Combat Swarm Drones for Navy's P-8i fleet, accessed August 24, 2025, <https://idrw.org/drdo-develops-air-launched-anti-radiation-combat-swarm-drones-for-navys-p-8i-fleet/>
- 18) How Drones Are Revolutionizing Search and Rescue - IEEE Public Safety Technology, accessed August 18, 2025, <https://publicsafety.ieee.org/topics/how-drones-are-revolutionizing-search-and-rescue/>
- 19) Increasing Operational Resiliency of UAV Swarms: An Agent-Focused Search and Rescue Framework - Frontiers Publishing Partnerships, accessed August 18, 2025, <https://www.frontierspartnerships.org/journals/aerospace-research-communications/articles/10.3389/arc.2023.12420/full>
- 20) Esp32 Development Board Price Starting From Rs 650/Pc. Find Verified Sellers in Ahmedabad - JdMart - Justdial, accessed August 18, 2025, <https://www.justdial.com/jdmart/Ahmedabad/esp32-development-board/jdm-1595478-ent-6-30744933>
- 21) Recognising hand gestures using Arduino and TensorFlow Lite - GitHub, accessed August 18, 2025, https://github.com/Hanqaqa/Arduino_Tensorflow_Gesture_Recognition
- 22) Multi-Agent Pathfinding with Real-Time Heuristic Search, accessed August 24, 2025, <https://yeoh-lab.wustl.edu/assets/pdf/cig-SigurdsonB0HK18.pdf>
- 23) Optimized Decentralized Swarm Communication Algorithms for Efficient Task Allocation and Power Consumption in Swarm Robotics - ResearchGate, accessed August 24, 2025, https://www.researchgate.net/publication/380140572_Optimized_Decentralized_Swarm_Communication_Algorithms_for_Efficient_Task_Allocation_and_Power_Consumption_in_Swarm_Robotics
- 24) Keyword Spotting on ESP32-S3 with INMP441 and MAX7219 - Hackster.io, accessed August 24, 2025, <https://www.hackster.io/amy/keyword-spotting-on-esp32-s3-with-inmp441-and-max7219-c3de33>
- 25) INMP441 MEMS High Precision Omnidirectional Microphone Module I2S, accessed August 24, 2025, <https://www.indianhobbycenter.com/products/inmp441-mems-high-precision-omnidirectional-microphone-module-i2s>
- 26) Category: Fun Friday | Bitcraze, accessed August 18, 2025, <https://www.bitcraze.io/category/fun-friday/>



I agree to file a patent of the proposed robot in ROBOFEST-GUJARAT 5.0 from the PIC Cell of GUJCOST.

-X-X-X-



Dhirubhai Ambani University

Formerly known as

Dhirubhai Ambani Institute of Information and Communication Technology

Mentor Undertaking Format

I Dr. Pankaj Kumar (Mentor) on behalf of my team authorized to give undertaking that on selection of our team at Level 1 (Ideation Stage), we assure and commit to participate in the subsequent levels and to submit Level 2 (Proof of Concept) and Level 3 (Proto type) robot as per guidelines of GUJCOST without fail, otherwise GUJCOST will take necessary action to recover the fund if disbursed for any stage.

Sign of Mentor.....

Sign of Director of Host Institute.....

Stamp of Institute/University

T. Bandyopadhyay
Director General
DAU, Gandhinagar





Formerly known as
Dhirubhai Ambani Institute of Information and Communication Technology

ENDORSEMENT FROM THE HEAD OF INSTITUTION

ROBOT CATEGORY: Aerial Robotics-Minefield Navigation Challenges

1. Certified that the Institute welcomes participation of **Dr. Pankaj Kumar** as the Mentor of the following students to participate in the ROBOFEST Gujarat 5.0 Competition:
 1. Rudra Bhatt
 2. Hari Sharma
 3. Vyom Patel
 4. Aum Bavarva
2. The basic facilities and such other Administrative facilities as per the need of the robot category, will be extended to the team throughout the duration of the project.
3. Institute assumes to undertake the financial and other management responsibilities of the project.

Date: 26-08-2025

Place: Gandhinagar

Signature & Seal of Head of Institute

T. Bandyopadhyay
Director General
DAU, Gandhinagar

